A-A-50735
15 December 1987

COMMERCIAL ITEM DESCRIPTION

TEST SET, MICROPROCESSOR BOARD

The General Services Administration has authorized the use of this Commercial Item Description.

Scope. This Commercial Item Description (CID), describes a test set, microprocessor board. This CID is meant as a minimum requirement for the test set, microprocessor board in which only those manufacturers meet or surpass the following requirements are qualified per this CID.

Classification. Equipment covered by this CID may be commercially available equipment modified to the extent necessary to meet the following description. The equipment shall be type III, class 6, style E and color R in accordance with the latest MIL-T-28800. This CID makes provision for the bid sample testing in accordance with the latest MIL-T-28800.

Applicable documents.

Specifications, and standards. The following specifications and standards, form a part of this CID to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-T-28800 Test Equipment for use with Electrical and Electronic Equipment, General Specification for (the latest edition)

(Copies of specifications and standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.

AMSC N/A FSC 6625

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Other publications. The following document forms a part of this CID to the extent specified herein. Unless otherwise specified, the issue of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI-C62.41-1980, 2.5KV, Category A

(Application for copies of ANSI publications should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

Order of precedence. In the event of a conflict between the text of this CID and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this CID shall take precedence. Nothing in this CID however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Salient characteristics. The equipment shall be capable of operation within the accuracies, limits, and specifications herein.

1. Safety. The equipment shall be UL (Underwriters Laboratories) listed and approved and or shall comply with the Type III safety requirements of the latest MIL-T-28800.

a. Leakage current protection. Protection shall be provided from leakage current in excess of 5 milliamperes (mA) rms (AC and DC) from any accessible conductive parts of equipment (including control shafts with knobs broken or removed and recessed calibration or adjustment controls) to either pole of the power source for any position of the power switch(es).

b. Radio frequency interference (RFI) capacitors. Any capacitor used in RFI filters shall be rated for at least four(4) times the operating voltage.

c. Type III ventilation openings.

(1) Ventilation openings shall be designed to prevent accidental contact with live electrical circuits and mechanical moving parts.

(2) For ventilation openings shall incorporate small mesh screening or similar design to prevent the expulsion of hot electronic particles due to failure of electronic devices such as capacitors.

2. Parts, materials, and processes. Except for marking, the test set, microprocessor board shall comply with the Type III parts, materials, and process requirements of MIL-T-28800. Nonstandard parts approval requests shall be in accordance with the latest MIL-T-28800. Front panel marking shall be in accordance with the latest MIL-T-28800.

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a. Restricted material. The equipment shall comply with the restricted material requirements of the latest MIL-T-28800, including the requirement for mercury and radioactive materials, and shall contain no combination of materials which cause deterioration of any material contained in the equipment due to the effects of outgassing.

3. Design and construction. Unless otherwise specified herein, the equipment shall comply with the type III design and construction requirements of the latest MIL-T-28800. The equipment shall be of solid-state construction. Encapsulation and embedment (potting) of subassemblies may be used only for high voltage circuits where construction is susceptible to leakage or corona, indicators and associated controls shall be in close proximity to each other, all potentiometer controls shall have at least a 270 degree rotation range (4.71 radians).

a. Type of construction. Construction of the test set microprocessor board shall be of a test set or system design.

b. Internal cooling. The equipment shall be capable of operating continuously for twenty four(24) hours at 35°C. If forced air is used, it shall be as specified in MIL-T-28800.

(1) Forced air cooling motor. DC brushless motors shall be utilized for the forced air cooling motor in the equipment.

(2) Air filter. Equipment utilizing a cooling scheme which exhausts air from the equipment through the cooling fan shall not require an externally accessible air filter.

(3) Thermal switch. A thermal switch shall be provided which will interrupt circuit power to the test set, microprocessor board when the internal temperature rises to the degree of possible damage to the electronic components. A minimum of 10°C hysteresis drop from this tripping temperature must occur before circuit power is again applied.

4. Electrical power sources. The equipment shall operate from nominal commercial, military, and shipboard power sources of 100, 120, 220 and 240 volts ac (+10% at line frequencies of 50 Hz, 60 Hz +10%.

a. Electrical power connectors. The equipment shall operate with a ground at the primary source or either side of the single phase input.

b. Power consumption. Maximum power consumption shall not exceed 300 watts.

c. Fuses. Fuse(s) shall be provided, on the exterior rear panel for replacement without entering the enclosure. The fuse(s) shall automatically disconnect the equipment from the power source in the event of excessive current requirements. Exceptions will be accepted for disk drives and printers.

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d. Surge and transient voltage protection. The test set microprocessor board shall contain surge and transient voltage protection on internal of the enclosure. The transient voltage protection circuit shall be so designed as to prevent electronic device damage within the system per type III solid state tolerance bands of MIL-T-28800, the surge voltage protection circuit shall conform to ANSI/IEEE Std. C62.41-1980, 2.5KV, category A. The surge protection circuit shall drain excess power due to surges directly to the ground conductor for personnel protection. Normal operation of the test set microprocessor board may be inhibited and false operational signals may be generated under input voltage transients greater than ±20 percent and frequency transient conditions greater than ±5 percent from within the nominal type III steady-state tolerance bands of MIL-T-28800, damage (other than fuse failure) to the test set microprocessor board shall not occur during voltage transient conditions greater than ±20 percent and frequency transient conditions greater than ±5 percent within the maximum limits (30 percent voltage transients and 10 percent frequency transients) specified in MIL-T-28800. Normal operation of the test set microprocessor board shall be automatically resumed when the transient has passed and any blown fuses are replaced. Blown fuses shall be externally accessible. Testing shall be in accordance with MIL-T-28800.

5. Dimensions and weight. The size and weight of the equipment shall be consistent with current commercial capabilities, refer to paragraphs 5a and 5b of this CID.

a. Equipment dimensions. The nominal dimensions for the test set, microprocessor board component level should be approximately 5 inches height x 16 inches wide x 14 inches deep.

b. Equipment weight. The component level weight including all accessories (excluding manual), shall not exceed 34 lbs.

6. Enclosure requirements. Unless otherwise specified herein, the equipment shall comply with the Style E enclosure requirements of MIL-T-28800.

a. Rackmount. Rackmount capability is not required on this tester.

b. Tilt bail handle. A tilt-bail handle is not required on this tester.

7. Marking and identification.

a. Marking. The visible surface adjacent to panel facilities such as variable controls, indicators, jacks, connectors and fuse holder(s), shall be marked with a suitable word, phrase, or abbreviation thereof, indicating the purpose or use of the part.

(1) Panel marking. The front panel marking shall comply with the type III parts, materials and processes panel marking requirements of MIL-T-28800, however, marking identification processes other than the type III requirements specified in MIL-T-28800 may be used, provided the marking and identification processes are capable of passing the front panel marking test when tested in accordance with this CID.

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b. Items to be marked.

(1) Volume and weight (including accessories) shall be marked on three external surfaces of the enclosure, if common enclosure is utilized.

(2) The type designation of the AN nomenclature of the test set, microprocessor board is not required.

8. Environmental requirements.

a. Temperature requirements. Unless otherwise specified herein, the equipment shall meet its specified performance characteristics and other requirements when subjected to the class 6 temperature requirements specified by MIL-T-28800 and as listed below.

(1) Non-operating temperature. The equipment shall meet its performance and accuracy requirements in an non-operating environment of -10°C to +50°C.

(2) Operating temperature. The equipment shall meet its performance and accuracy requirements in an operating environment of 10°C to 35°C.

b. Humidity requirements. The equipment shall meet its required performance and accuracy requirements under humidity requirements for class 6 equipment as specified by MIL-T-28800.

c. Altitude requirements. The equipment shall meet its required performance and accuracy requirements under the altitude requirements for class 6 equipment as specified by MIL-T-28800.

d. Vibration requirements. The equipment shall meet the class 6 vibration requirements as specified by MIL-T-28800.

e. Mechanical shock requirements.

(1) Shock, functional. The equipment shall meet the class 6 shock, pulse requirement as specified by MIL-T-28800.

(2) Transit drop. The equipment shall meet transit drop requirement as specified by MIL-T-28800.

(3) Bench handling. The equipment shall meet the requirements as specified by MIL-T-28800.

(4) Shock, high impact. The shock, high impact is not required for this tester.

f. Electromagnetic interference (EMI). The equipment shall comply with standard commercial EMI practices.

g. Magnetic environment. The equipment shall comply with the magnetic environment requirements of MIL-T-28800.

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9. Reliability. Reliability verification testing, shall be in accordance with the type III, class 6 environmental conditions of MIL-T-28800.

a. Mean time between failures (MTBF). The equipment mean time between failures (MTBF) shall be 2,000 hours.

b. Pre-conditioning requirement. Each deliverable equipment shall be capable of being subjected to the environmental screening effects of burn-in and thermal cycling, with thermal cycling being accomplished in the sample testing phase.

(1) BURN-IN. Burn-in shall expose the equipment to extended periods of elevated temperatures, (40°C for a minimum of 40 hours). Each equipment which fails during burn-in shall be repaired and fully retested, until failure free. Failures which occur during the burn-in test shall be noted and reported but shall not count toward the establishment of MTBF. Prior to and following burn-in, a satisfactory operation test (SOT) shall be performed. A satisfactory operation check (SOC) shall be performed conducted at selected times during the burn-in test.

(2) Thermal cycling. Thermal cycling shall expose the equipment to hot and cold cycles for class 6 equipment per MIL-T-28800. Each equipment which fails during thermal cycling shall be repaired and full cycling test shall be noted and reported but shall not count toward the cycling, a satisfactory operation check (SOC) shall be conducted during the last two upper and lower temperature limits.

10. Calibration and maintenance adjustments. The design of the syster shall provide for readily accessible calibration adjustments and maintenance adjustments. These adjustments shall be provided by variable value components which are adjustable by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules.

a Maintainability. The equipment shall meet the type III maintainability requirements specified by MIL-T-28800.

11. Interchangeability. The equipment shall meet the type III interchangeability requirements specified by MIL-T-28800.

12. Digital interface. The equipment shall meet the digital interface requirements per MIL-T-28800.

13. Performance characteristics. The test set microprocessor board shall test and learn the following microprocessor, and associated RAMs, ROMs, I/O read-write registers: 1802, 1804, 1805, 1806, Z80, 6502, 6800, 6802, 6802NS, 6808, 6809, 6809E, 8035, 8039, 8040, 8041, 8041A, 8042, 8048, 8049, 8050, 8741, 8741A, 8742, 8748, 8749, 8080, 8085, 8086, 8088, 9900, 68000, 80180, 80280, 80286, 80386, 68008, 68010, 68020 and 68032.

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a. Controller. The controller shall be an operator interface keyboard, a minimum 32-character operator message center display, RS-232C interface port, interface port for logic probe, interface port for microprocessor and microprocessor socket.

(1) Controller functions. The controller shall contain but not be limited to the following functions:

(2) Self test. The controller shall perform a complete self test and system test on power up.

(3) Learn. The controller shall self learn by utilizing a known good microprocessor and board store the results into memory for immediate testing or save learned data on cassette tape.

(4) View. The controller will display all learn results, including all addresses and data locations, comparison of known architects and allow manual entry of RAM, ROM, I/O information.

(5) Functional tests. The controller shall perform the following functional test.

(a) Electrical integrity of the busses.

(b) Computes ROM signatures and compares to a stored reference signature.

(c) Checks, verifies, detects RAM read-write ability address decoding and data line shorts beyond the bus buffers.

(d) Checks read-write ability of all I/O registers.

(e) In auto the controller will run all test automatically microprocessors.

(f) Monitors power supply.

(g) Tests the microprocessors clock.

b. Microprocessor interface. The test set, microprocessor board shall come with interface cable and connections of the microprocessor board to a 8, 16, or 32 bit microprocessor to the microprocessor test set. The cable shall be a minimum of 2M in length. The interconnect shall contain protection for invertantly connecting the microprocessor board, microprocessor or microprocessor test set backwards with no damage.

c. The test set, microprocessor shall come with mag tape cassette either contained within the controller or separate of the controller with interconnect cable. The cassette shall store all learned, programmed, prompting and testing result via the controller operator keyboard operation.

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d. Test speed. The test speed will be depended on the microprocessor's clock.

e. Probe The test set shall come with a logic probe and a 2M interface cable. The probe shall be used as a single point measurement-logic state, signature, event count via controller. The probe may be used asynchronous or synchronized to the microprocessor address or data. The probe shall be utilized as a hi/lo pulse stimulus.

f. Manuals. The test set, microprocessor board shall come with two(2) sets of operating, programming and instruction manuals.

14. Calibration interval. After calibration, the equipment shall meet each performance requirement within the tolerances specified for a period of at least 9 months, based on an operating time of 1500 hours.

Preservation, packaging, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

Certification. The contractor shall certify that the product offered meets the salient characteristics of this description and that the product conforms to the producer's own drawings, specifications, standards and quality assurance practices and is the same product offered for sale in the commercial marketplace. Small businesses that cannot offer products with established commercial market acceptability, may compete by submission of bid samples. (The Government reserves the rights to require proof of such conformance before first delivery and thereafter as may be otherwise provided for under the provisions of the contact..

Known acceptable products. The following product(s) are known to have previously met the requirements of this CID and need not resubmit bid samples or certification. All other manufacturers must meet the requirements of this CID and submit to bid testing and certification as required.

Manufacturer	Model	NSN
Fluke	9010A/AF	

Custodian:
Air Force - 99
Navy - EC
DLA - ES

Preparing activity.
Air Force - 82

Reviewing activity
Army -CR

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